REHOTEQ AI EDUCATION CURRICULUM FOR NIGERIAN SCHOOLS

Master Framework and Implementation Blueprint
Draft 1.0 | October 2026 | Primary 1–SSS 3

Proposed curriculum developed by Rehoteq Technologies

Status and use

This document is a proposed curriculum framework for consultation, school pilots and further development. It is not an approved national syllabus, official examination syllabus or recognised qualification. Before formal adoption, Rehoteq should map the proposal against the current NERDC curricula, confirm subject-offering rules with relevant authorities, and obtain any approvals required for the chosen delivery model.

1. Vision and rationale

Vision: equip Nigerian learners to understand artificial intelligence, use it safely and creatively, and develop solutions to meaningful problems in their schools and communities.

The programme builds on the uploaded Rehoteq secondary scheme (JSS 1–SSS 3), retaining its practical orientation, Nigerian examples, three-term school rhythm, two-period weekly model, programming pathway, data literacy, machine learning, ethics and capstone projects. Primary 1–6 is added as a proposed age-appropriate foundation.

2. Curriculum principles

- **Age-appropriate progression:** Move from play, patterns and simple rules in early primary to programming, data and AI system design in secondary.
- **Human-centred AI:** AI supports learning and problem-solving; it does not replace teacher judgement, learner effort or human responsibility.
- **Nigerian relevance:** Use familiar contexts such as farming, markets, local languages, transport, school administration, public health information and environmental care.
- **Equity and low-resource delivery:** Every unit should have an offline alternative using paper, cards, group discussion or teacher demonstration.
- **Practice before abstraction:** Each term includes practical tasks, reflection and a small project.
- **Safety and privacy by design:** Teach consent, age-appropriate use, data minimisation, misinformation detection and safe digital conduct throughout.
- **Evidence of learning:** Assess explanation, process, testing, reflection and practical outcomes—not memorisation alone.
- **Continuous review:** Update tools and examples while keeping core concepts stable.

3. Proposed graduate profile

- Explain AI in accessible language and distinguish AI-enabled systems from ordinary rule-based software.
- Ask useful questions, break problems into steps, and reason with data.

- Use digital and AI tools appropriately, checking outputs against trustworthy evidence.
- Recognise privacy, security, fairness, misinformation and environmental concerns.
- Create and present age-appropriate artefacts, programs, data analyses or AI prototypes.
- Work collaboratively, document decisions, accept feedback and communicate limitations.
- Connect AI skills to Nigerian community needs, further study and career pathways.

4. Competency strands

- **A. Digital foundations —** Devices, files, operating systems, networks, online safety, accessibility and responsible digital citizenship.
- **B. Computational thinking —** Patterns, decomposition, algorithms, flowcharts, debugging, block coding and text-based programming.
- **C. Data literacy —** Collecting, organising, cleaning, representing, interpreting and protecting data.
- **D. AI understanding —** AI examples, rule-based systems, machine learning, training data, inference, model limits and evaluation.
- **E. Generative AI literacy —** Prompts, output verification, hallucinations, attribution, academic honesty, copyright and age-appropriate use.
- **F. Ethics, society and safety —** Human agency, privacy, fairness, inclusion, security, misinformation, accountability and sustainability.
- **G. Design and creation —** Problem definition, user needs, prototyping, testing, iteration, documentation and presentation.
- **H. Enterprise and careers —** Teamwork, project planning, portfolios, entrepreneurship, further study and local technology ecosystems.

5. Learning progression by class

Class	End-of-year learning focus (proposed)	Illustrative evidence/project
Primary 1	Notice patterns; follow instructions; identify smart devices; learn basic device care and privacy rules.	Pattern cards; instruction sequence; draw a helpful smart device.
Primary 2	Sort and classify objects; understand inputs and outputs; practise safe guided digital activities.	Sorting game; input/output role-play; picture story.
Primary 3	Use simple decision rules; distinguish people from machines; collect small class data.	Rule-based guessing game; tally chart; AI around us poster.
Primary 4	Understand data examples and patterns; use block-based coding; discuss why a machine may make mistakes.	Scratch animation; classify local objects; error-spotting activity.
Primary 5	Explain that AI systems use examples/data; compare AI suggestions with evidence; practise responsible prompting with teacher guidance.	Small labelled dataset; prompt-and-check worksheet; local problem poster.
Primary 6	Plan a simple project, explain data and rules, test an age-appropriate AI demo and present limitations.	Community helper prototype; class presentation; AI safety pledge.
JSS 1	Digital foundations, AI in everyday Nigerian services, classification, data collection, Scratch sequences, loops, variables and conditions.	Class survey dashboard or Scratch learning game.

JSS 2	Python basics, spreadsheets, simple chatbots, recommendation systems, data quality, introductory model training and testing.	School FAQ bot or local image/sound classifier.
JSS 3	Algorithms, Python functions and data structures, web forms, misinformation checks, AI bias, project documentation and capstone planning.	Data-informed school/community project with a tested prototype.
SSS 1	Python fundamentals, data analysis, statistics, supervised learning, model evaluation, web foundations and databases.	Train and evaluate a simple classifier; publish a portfolio page.
SSS 2	Data cleaning, regression and classification, visualisation storytelling, NLP and computer vision concepts, APIs and application development.	AI-assisted app prototype with evaluation and ethical risk review.
SSS 3	Advanced project cycle, model evaluation, documentation, deployment concepts, responsible AI, career pathways and portfolio presentation.	Capstone, technical report, presentation and viva.

6. Three-term scope and sequence

Planning assumption: 13 weeks per term and two periods per week, following the uploaded Rehoteq secondary scheme. The 13-week structure is a planning template, not a claim that every Nigerian school uses identical dates. Schools should adapt calendars and examination weeks to their approved timetable. For primary classes, each period should be short, play-based and appropriate to the learners' age.

Primary 1

- Term 1: patterns and sequences; instructions; technology around us; device care; privacy and trusted adults; unplugged project.
- Term 2: sorting and grouping; inputs and outputs; simple if/then rules; guided digital drawing/storytelling; check-and-share routines.
- Term 3: people and machines; helpful technology; simple data tallies; review; showcase a picture story or classroom helper idea.

Primary 2

- Term 1: classify by attributes; step-by-step instructions; digital devices and safe handling; simple algorithms.
- Term 2: inputs/outputs; rule-based games; patterns in pictures and sounds; guided creative tools; safe sharing.
- Term 3: collect and represent small datasets; compare human and machine decisions; make a classroom technology poster/project.

Primary 3

- Term 1: data as information; sorting rules; sequences and debugging; technology in community services.
- Term 2: block coding foundations; events and repetition; simple decision trees; responsible use and privacy.
- Term 3: AI examples and limits; verify a generated or suggested answer with teacher help; community problem mini-project.

Primary 4

- Term 1: digital literacy; data tables; patterns and classification; algorithms and flowcharts.
- Term 2: block coding with events, loops and conditions; inputs/outputs; debugging; data visualisation basics.
- Term 3: how AI uses examples; errors and bias; online safety; design and present a simple interactive project.

Primary 5

- Term 1: data quality; labelled examples; rules versus learning from examples; block coding and decomposition.
- Term 2: guided AI demonstrations; prompts and verification; image/text classification concepts; charts and evidence.
- Term 3: privacy, fairness and misinformation; project planning; testing and reflection; community-focused showcase.

Primary 6

- Term 1: computational thinking; data collection; spreadsheet basics; AI systems and their limits.
- Term 2: block-based project; model training demonstration where safe and available; output checking; accessibility and inclusion.
- Term 3: capstone planning and presentation; AI ethics; digital portfolio; transition to JSS.

JSS 1

- Term 1: computer and society; hardware/operating systems; word processing and internet safety; what AI is; Nigerian AI examples; classification and surveys.
- Term 2: spreadsheets and charts; Scratch interface, sequences, loops, variables and conditions; binary basics; project.
- Term 3: presentations; Scratch game; algorithms, flowcharts and pseudocode; AI ethics, digital citizenship and e-waste.

JSS 2

- Term 1: networks and search; HTML; Scratch events; data quality; Python first steps; AI recognition systems and recommendations; forms.
- Term 2: Python variables, conditionals, loops, lists and functions; Scratch-to-Python; rule-based chatbot; deepfakes and honesty.
- Term 3: databases and spreadsheets; machine-learning concepts; train/test a simple model; AI safety; careers and graphics project.

JSS 3

- Term 1: algorithms and efficiency; Python functions, lists and dictionaries; web forms; AI and society; fact-checking; survey/data project.
- Term 2: Python mini-projects; APIs; cloud concepts; version control; classification; data cleaning; bias; compare rule-based and AI chatbots.
- Term 3: adapt to BECE timetable; portfolio and capstone work when possible; Python, website and AI project; SSS pathway and careers.

SSS 1

- Term 1: data representation; problem solving; Python fundamentals and control flow; functions; spreadsheets; data-science process; AI/ML/deep learning overview.
- Term 2: Python collections and files; pandas; visualisation; descriptive statistics; labelled data; introductory classifier; consent and data rights.
- Term 3: HTML/CSS; JavaScript basics; portfolio website; cloud concepts; SQL and data modelling; AI service demonstrations; presentation.

SSS 2

- Term 1: responsive web design; JavaScript DOM; frontend/backend concepts; Flask; APIs; databases; project planning.
- Term 2: advanced data cleaning and analysis; visual storytelling; regression and classification; model evaluation; cloud deployment concepts.
- Term 3: NLP and computer vision; chatbot development; AI law/ethics; productivity with AI; entrepreneurship; capstone planning and prototype.

SSS 3

- Term 1: syllabus and skills audit; advanced Python concepts; capstone selection; ML project cycle; report writing; targeted revision.
- Term 2: capstone build, testing, documentation, deployment where feasible, presentation practice, mock assessment and gap remediation.
- Term 3: adapt to official examination timetable; revision and candidate support; portfolio completion; capstone showcase and post-school pathways.

7. Standard 13-week term template

Week(s)	Purpose	Suggested approach
1	Diagnostic / recap	Find prior knowledge and set learning goals.
2–4	Concept introduction	Teach core ideas with examples, discussion and guided practice.
5–7	Skill building	Practice, coding/data exercises, debugging and peer collaboration.
8–9	Applied investigation	Use a Nigerian context; check sources and consider safety/ethics.
10	Project and revision	Finish a small project and revise key concepts.
11	Assessment	Written, oral and/or practical assessment.
12	Feedback and remediation	Return feedback, correct misconceptions and support learners.
13	Reflection / records / closure	Portfolio update, results and school calendar activities.

8. Suggested weekly lesson structure

- Period 1 — Teach and model: learning objective, vocabulary, short explanation, teacher demonstration and guided questioning.
- Period 2 — Practise and apply: paired/group activity, practical task or project milestone, learner explanation and exit ticket.
- For primary learners, use short activity cycles, movement, pictures and oral explanation; avoid requiring personal accounts or unsupervised public AI tools.
- Where a school has only one period, combine the key explanation with a small practical task and carry project work forward.

9. Teaching and learning methods

- Unplugged activities: paper sorting, role-play, decision trees, flowcharts and card-based model simulations.
- Demonstration and guided practice: teacher shows a process, learners repeat it, then adapt it.

- Project-based learning: small teams address a defined problem and document choices, tests and limitations.
- Inquiry and discussion: learners ask questions, compare evidence and explain why an AI output may be wrong.
- Peer learning: learners test each other's work using respectful, structured feedback.
- Localised examples: market prices, crop observations, school library records, transport patterns, local language learning resources and environmental monitoring.
- Inclusive practice: provide non-screen options, accessible materials, multiple ways to demonstrate learning and avoid collecting sensitive learner data.

10. Assessment framework

Suggested assessment mix for pilot testing; schools and authorities may require adjustments.

Assessment area	Proposed weight	Evidence
Concept understanding	20%	Short quizzes, oral explanations, diagrams and vocabulary.
Practical skills	25%	Code, data work, safe tool use, debugging or model testing.
Project / portfolio	25%	A working artefact, logbook, screenshots or offline equivalent.
Reasoning and verification	15%	Source checking, test cases, interpretation and limitation analysis.
Ethics and collaboration	15%	Privacy decisions, fairness reflection, teamwork and responsible attribution.

Use a common 4-level rubric: 4 = independent and well-justified; 3 = meets the objective with minor support; 2 = partially meets the objective and needs guidance; 1 = beginning / needs substantial support. For younger learners, prioritise observation, oral explanation and artefacts over formal written tests.

11. Safety, privacy and responsible use policy

- Follow applicable Nigerian data-protection requirements and school safeguarding policies; obtain appropriate consent where needed.
- Do not enter student names, contact details, grades, photos, health information or other personal/sensitive data into public AI tools.
- Use age-appropriate tools and follow each tool's age requirements and school policy; teacher supervision is required for demonstrations.
- Treat AI outputs as suggestions, not facts. Verify important claims using reliable sources, textbooks, teachers or primary evidence.
- Do not use AI to impersonate, bully, create non-consensual images, fabricate evidence or deceive in assessments.
- Teach learners to disclose meaningful AI assistance and remain responsible for submitted work.
- Avoid biometric or facial-recognition projects involving identifiable students; use synthetic or consented, non-sensitive data instead.
- Review tool terms, data storage, advertisements, costs, accessibility and offline alternatives before classroom adoption.

12. Resources and low-resource delivery

Level	Resources
Minimum	Teacher guide, printed activity cards,

	board/markers, paper, locally collected non-sensitive datasets and assessment rubrics.
Basic digital	Shared computers or tablets, offline coding tools where available, projector or teacher demonstration device, local file storage.
Enhanced	Reliable internet, supervised accounts where permitted, AI demonstration tools, Python environment, spreadsheets, data visualisation and version-control tools for older learners.

No learning outcome should depend solely on paid subscriptions, continuous broadband, a personal smartphone or a learner creating an account on a public platform.

13. Teacher development pathway

1. Module 1: AI foundations, common terminology, strengths and limitations.
2. Module 2: Teaching computational thinking and age-appropriate programming.
3. Module 3: Data literacy, model demonstrations, testing and error analysis.
4. Module 4: Responsible AI, privacy, safeguarding, misinformation and academic integrity.
5. Module 5: Project-based teaching, differentiated instruction and practical assessment.
6. Module 6: Lesson planning, reflective practice, classroom implementation and peer review.

Teacher readiness should be assessed through a short knowledge check, a model lesson, a sample learner activity and a reflection on safety. Certification should only be described as a Rehoteq completion certificate unless formally accredited by an authorised body.

14. Implementation and development plan

7. **Phase 1 — Governance and validation (Weeks 1–4):** Appoint curriculum lead and reviewers; confirm target school types; obtain current NERDC documents; map subject and timetable constraints; create a curriculum advisory group.
8. **Phase 2 — Detailed curriculum writing (Weeks 5–12):** Develop class-level learning outcomes, term schemes, vocabulary, lesson plans, practical activities, answer keys and rubrics; identify offline alternatives.
9. **Phase 3 — Resource and teacher package (Weeks 9–14):** Create teacher handbook, student workbooks, slide templates, project bank, safety policy and teacher training modules.
10. **Phase 4 — Pilot preparation (Weeks 13–16):** Select pilot schools representing different resource levels; baseline learners; train teachers; check devices, consent and safeguarding arrangements.
11. **Phase 5 — Pilot delivery (One school term):** Teach selected classes; observe lessons; collect anonymised assessment results, teacher feedback and learner work; record cost and connectivity constraints.
12. **Phase 6 — Review and revision (4–6 weeks after pilot):** Analyse learning evidence; revise overloaded or unclear units; check accessibility and cultural relevance; publish versioned curriculum.
13. **Phase 7 — Wider adoption and authority engagement:** Prepare stakeholder brief, implementation budget, quality assurance evidence and formal submission materials if required.

15. Pilot evaluation measures

- Learner progress: pre/post measures of key concepts and practical competencies.
- Project quality: proportion of learners who can explain, test and improve a project.
- Responsible use: ability to identify privacy risks, verify outputs and explain limitations.
- Teacher confidence: baseline and post-training self-assessment plus observed teaching practice.
- Equity: participation and outcomes across device access, gender, language and learner support needs, using privacy-protective methods.
- Feasibility: teaching time, device availability, connectivity, material costs and lesson completion.

- Adoption: school leadership feedback, timetable fit and willingness to continue.

16. Governance and quality assurance

- Curriculum lead: owns scope, progression and version control.
- Subject reviewers: review computing, AI, pedagogy, child safeguarding, data protection and Nigerian context.
- Teacher reviewers: check lesson clarity, realistic timing, materials and assessment load.
- Pilot schools: provide structured feedback and anonymised evidence.
- External alignment review: confirm mapping to current NERDC documents and relevant examination requirements.
- Version control: every edition records date, author/reviewer roles, changes, evidence and approval status.

17. Risks and mitigations

Risk	Mitigation
Subject/timetable approval	Do not claim national approval; verify current subject rules and use permitted school-based delivery during development.
Unequal device access	Design offline equivalents and shared-device/group activities.
Rapid tool changes	Teach durable concepts and review tools each school year.
Student privacy	Use synthetic or anonymised data, school policies and age-appropriate supervised tools.
Teacher readiness	Provide staged training, exemplar lessons and mentoring.
Overloaded syllabus	Prioritise core outcomes; reserve extension topics for schools with more time/resources.
Exam misalignment	Maintain a separate, verified mapping to current examination syllabuses; never assume AI projects replace required exam content.

18. Required deliverables for the complete package

- Master curriculum framework and competency map.
- Primary 1–6 scheme of work, lesson plans and student activity books.
- JSS 1–3 revised scheme of work, teacher notes, practicals and assessment materials.
- SSS 1–3 detailed scheme of work, technical labs, capstone guidance and portfolio rubric.
- Teacher training handbook and facilitator slides.
- AI safety, privacy, safeguarding and academic integrity policy.
- Practical project bank with offline and digital options.
- Question bank, marking guides, practical assessment rubrics and sample report cards.
- Resource inventory, indicative cost model and implementation guide.
- NERDC/examination mapping matrix, stakeholder proposal and pilot evaluation report.

19. Immediate next actions

14. Obtain and archive the latest official NERDC curriculum documents for Primary 1–6, JSS 1–3 and SSS 1–3.
15. Decide the initial pilot footprint: recommended start is Primary 4–6 plus JSS 1, while developing the full scope in parallel.
16. Set the intended weekly teaching time; this framework assumes two periods per week as a planning baseline.

17. Recruit reviewers from primary teaching, computer science/AI, curriculum development, safeguarding and school leadership.
18. Write and test one complete sample unit before scaling to all 12 classes.
19. Agree on a transparent status label: 'Rehoteq proposed curriculum — pilot edition' until formal approval is secured.

20. Reference points

User-provided source: Rehoteq AI Curriculum: Secondary (JSS 1 to SSS 3), Edition 2.0, Scheme of Work, Session 2026/2027.

NERDC: https://www.nerdc.gov.ng/content_manager/new_curriculum_home.html

NERDC Senior Secondary Curriculum: https://nerdc.gov.ng/content_manager/new_senior_curriculum_home.html

NERDC e-Curriculum portal: https://www.nerdc.org.ng/CurriculumStructure.aspx

UNESCO, AI competency framework for students (2024; updated 2026): https://www.unesco.org/en/articles/ai-competency-framework-students

Appendix A. Sample lesson blueprint

Class / term	Primary 5, Term 2 (illustrative)
Topic	How AI learns from examples
Duration	Two periods in one week
Learning outcomes	Learners can explain that some AI systems use examples; identify a label and an example; explain one reason a classifier may make an error.
Key vocabulary	AI, data, example, label, prediction, error
Materials	Picture cards or paper cards; board; optional teacher-led image classifier demonstration.
Period 1	Starter: sort objects using a human-chosen rule. Discuss how examples help identify patterns. Model labels and predictions. Ask learners what could confuse the system.
Period 2	Teams label a small, non-sensitive set of pictures, test a simple rule or supervised demonstration, record mistakes and propose one improvement.
Assessment	Exit ticket: define example and label; explain one error; state one safety rule.
Offline option	Use printed cards and a human 'classifier' following a rule; compare results and discuss limitations.
Safety	Do not use identifiable student photos or upload personal information. Use classroom objects or synthetic images.
Extension	Compare two different sets of examples and explain why the results might differ.